ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY

NTA LEVEL 5 - TECHNICIAN CERTIFICATE

CONTINUOUS ASSESSMENT TEST - I - MAY-2026

CET 05204: COMPUTER ARCHITECTURE AND ASSEMBLY LANGUAGE

SEMESTER: II | YEAR: II

DURATION: 1 Hr 30 min | MAX. MARKS: 50

PART - A (Answer all the Questions)

Q1. What does a term BUS mean in computer architecture?

In Computer Architecture, a BUS is a communication system or pathway that transfers data, memory addresses, and control signals between computer components inside a computer, or between computers. It is essentially the internal "highway" of the motherboard.

Q2. Define the term register.

A register is a small, extremely fast memory location situated directly within the CPU. It is used to quickly store, accept, and transfer data and instructions that are being used immediately by the CPU.

Q3. What is a long form of PC in concept of registers?

The full form of PC is Program Counter. It is a register that holds the memory address of the next instruction to be fetched and executed by the CPU.

Q4. Give two disadvantages of Input-Output Processor.

Two disadvantages of an Input-Output Processor (IOP) are:
1. Increased Cost: Adding an independent IOP increases the overall hardware cost of the computer system.
2. Design Complexity: It adds significant complexity to the system hardware design and the operating system software, as it must coordinate communication between the main CPU and the IOP.

Q5. Mention three types of micro operations.

Three types of micro-operations are:
1. Register Transfer Micro-operations: Used to transfer binary information from one register to another.
2. Arithmetic Micro-operations: Used to perform basic arithmetic operations on numeric data stored in registers (e.g., addition, subtraction).
3. Logic Micro-operations: Used to perform bit manipulation operations on non-numeric data (e.g., AND, OR, XOR).

Q6. What is the function of IR?

IR stands for Instruction Register. Its primary function is to hold the current instruction that has been fetched from memory and is currently being decoded or executed by the CPU.

Q7. Write the DVD in full form?

The full form of DVD is Digital Versatile Disc (sometimes also referred to as Digital Video Disc).

PART - B (Answer any three Questions)

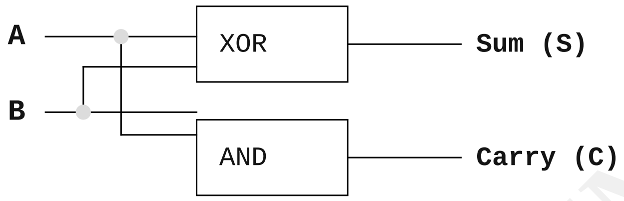
Q8. Explain briefly about half adder with logic table and logic circuit.

A Half Adder is a combinational logic circuit used to add two single-bit binary numbers (A and B). It produces two outputs: Sum and Carry.

Logic Table (Truth Table):

Input A	Input B	Sum (S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Logic Circuit for Half Adder:



SMART TIP: In the block diagram above, the XOR gate produces the Sum by adding A and B, while the AND gate produces the Carry out.

Q9. Give four features of an Input-Output Processor.

Four comprehensive features of an Input-Output Processor (IOP) are:

- 1. Independent Processing: It acts as an independent processor capable of fetching and executing its own specialized I/O instructions from memory, functioning almost like a secondary CPU dedicated entirely to I/O tasks.

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Four types of secondary memory are:

- Magnetic Disks: Use magnetic storage to store data on rotating platters. They offer high capacity and low cost. Example: Hard Disk Drive (HDD).

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Four important types of registers include:

- Program Counter (PC): Holds the address of the next instruction to be fetched from memory, dictating the execution flow.

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The two main types of Control Units are:

1. Hardwired Control Unit: It is implemented using physical electronic hardware components (logic gates, flip-flops, decoders) wired together to generate control signals. It is extremely fast but highly inflexible, making it difficult to modify or add new instructions once the hardware is manufactured.

2. Microprogrammed Control Unit: It uses small programs (microprograms or microcode) stored in a special, fast control memory to generate control signals. It is slightly slower than the hardwired approach but highly flexible, allowing engineers to easily update, add, or fix instructions by simply modifying the microcode.

Q13. Explain briefly the four types of Buses.

While the main System Bus is divided into three components, we generally recognize these four types based on function and scope:
1. Data Bus: Carries the actual data being transferred between components (e.g., between CPU and Memory).
2. Address Bus: Carries the memory address of the location where data needs to be written to or read from.
3. Control Bus: Carries control signals (like Read, Write, Interrupt) that coordinate and synchronize the activities of the entire system.
4. Expansion Bus (I/O Bus): Connects peripheral external devices to the computer system (e.g., USB bus, PCI bus).

PART - C (Answer any two Questions)

Q14. Discuss in detail the six types of registers.

Registers are high-speed storage locations within the CPU. Six primary types of registers are:

1. Accumulator (AC): The most frequently used register, primarily linked with the Arithmetic Logic Unit (ALU). It holds initial operands and stores the final results of arithmetic and logical operations.

2. Program Counter (PC): Crucial for program execution flow. It keeps track of the memory address of the next instruction to be fetched, automatically incrementing after each fetch.

3. Instruction Register (IR): Once an instruction is fetched from memory, it is temporarily stored in the IR so the Control Unit can decode and execute it.

4. Memory Address Register (MAR): Holds the specific memory address of the data or instruction that the CPU wants to read from or write to. It connects directly to the Address Bus.

5. Memory Data Register (MDR) / Memory Buffer Register (MBR): Acts as a buffer. It holds the actual data being read from memory or the data waiting to be written to memory. It connects to the Data Bus.

6. Data Register (DR): A general-purpose register used to hold temporary data (operands) required by the ALU during operations, facilitating data transfer between the CPU and memory.

Q15. Discuss in detail about control unit in detail.

The Control Unit (CU) is a vital component of the CPU, acting as the system's "manager." While it does not process data itself, it directs and coordinates all operations across the computer.

Key Functions of the Control Unit:
1. Generating Control Signals: It issues commands to the ALU, Memory, and I/O devices, telling them what operations to perform and when.
2. Instruction Decoding: It reads the instruction currently residing in the Instruction Register (IR) and translates it into a series of actionable steps.
3. Timing and Synchronization: It ensures that all parts of the computer work in harmony by utilizing a system clock to coordinate the exact timing of data transfers and operations.

Types of Control Units:

- Hardwired Control Unit: Built purely from hardware circuitry (logic gates). It offers the highest speed but lacks flexibility for updates.
- Microprogrammed Control Unit: Uses software logic (microcode) stored in a control memory. It is highly adaptable and easier to design, though marginally slower than the hardwired counterpart.

Q16. Discuss in detail about Asynchronous data transfer.

Asynchronous data transfer is a method of transmitting data between two independent units (e.g., a CPU and a slower I/O device) that do not share a common clock. Because they operate at different speeds, they require special communication protocols to ensure data is transferred correctly without loss.

To achieve this synchronization, two primary techniques are used:

1. Strobe Control: This involves a single control line. One unit (either the source or destination) sends a "Strobe pulse" to indicate that data is available on the bus or that it is ready to receive data. The main drawback is that the sender has no way of knowing for sure if the receiver successfully captured the data.

2. Handshaking: This is a more reliable, two-way communication protocol. It utilizes two control lines for mutual acknowledgment. For example, the sender places data on the bus and sends a "Data Valid" signal. The receiver reads the data and replies with a "Data Accepted" signal. The sender waits for this acknowledgment before clearing the bus. This guarantees safe data transfer regardless of the speed mismatch between the devices.

Q17. Discuss the components of CPU in detail.

The Central Processing Unit (CPU) is the brain of the computer. It consists of three primary components working together to process instructions:

1. Arithmetic Logic Unit (ALU):
The ALU performs all mathematical calculations (addition, subtraction, multiplication, division) and logical operations (AND, OR, NOT, value comparisons). The ALU acts strictly based on the signals it receives from the Control Unit.

2. Control Unit (CU):
The CU is the supervisor of the CPU. It manages the fetch-decode-execute cycle. It fetches instructions from memory, decodes them to understand what needs to be done, and generates specific control signals directing the ALU, memory, and registers to execute the task.

3. Registers:
Registers are extremely fast, temporary storage areas located directly inside the CPU. They hold data, addresses, and instructions currently being processed, significantly speeding up execution. Key registers include the Program Counter (PC) for tracking instruction sequences, the Accumulator (AC) for storing ALU results, and the Instruction Register (IR) for holding the current instruction.

SMART TIP: The CPU connects to external components (Main Memory and I/O devices) using the System Bus, which consists of the Data Bus, Address Bus, and Control Bus.